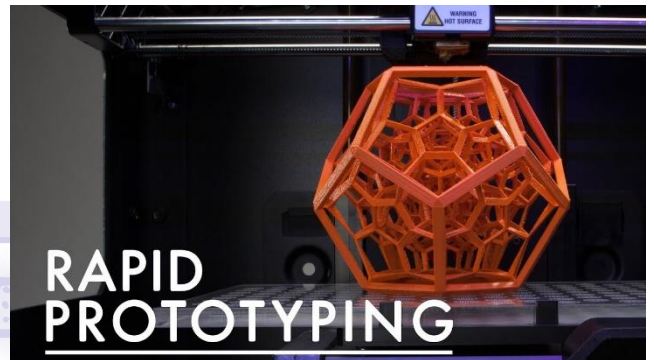
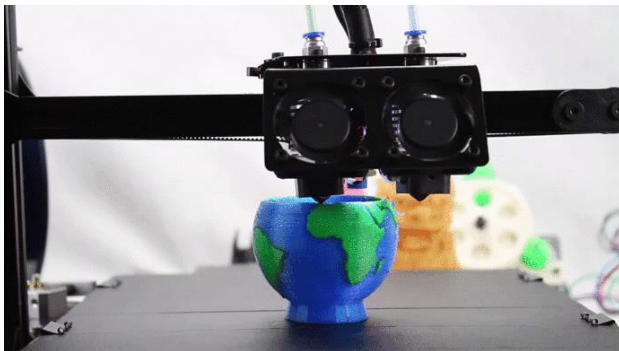
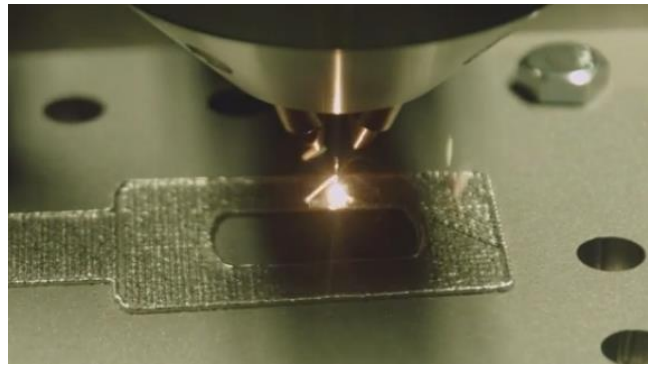
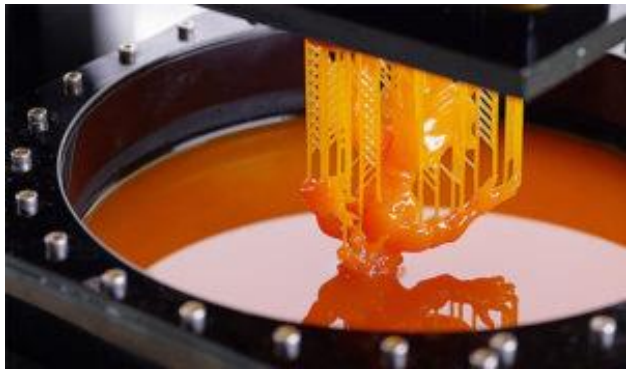


Introduction to Additive manufacturing



YANTRA

What Is Additive Manufacturing?

Additive manufacturing is the formalized term for what used to be called rapid prototyping and what is popularly called 3D Printing. The term rapid prototyping (RP) is used in a variety of industries to describe a process for rapidly creating a system or part representation before final release or commercialization. In other words, the emphasis is on creating something quickly and that the output is a prototype or basis model from which further models and eventually the final product will be derived.



Additive manufacturing is the formalized term for what used to be called rapid prototyping and what is popularly called 3D Printing. The term rapid prototyping (RP) is used in a variety of industries to describe a process for rapidly creating a system or part representation before final release or commercialization. The rapid prototyping term is inadequate and in particular does not effectively describe more recent applications of the technology. Many parts are in fact now directly manufactured in these machines, so it is not possible for us to label them as “prototypes.” Recently adopted ASTM consensus standards now use the term additive manufacturing



Prototype



Final product

What Are AM Parts Used for?

Initially, AM was used specifically to create visualization models for products as they were being developed. It is widely known that models can be much more helpful than drawings or renderings in fully understanding the intent of the designer when presenting the conceptual design. While drawings are quicker and easier to create, models are nearly always required in the end to fully validate the design.

Following this initial purpose of simple model making, AM technology has developed over time as materials, accuracy, and the overall quality of the output improved. Models were quickly employed to supply information about what is known as the “3 Fs” of Form, Fit, and Function. The initial models were used to help fully appreciate the shape and general purpose of a design (Form). Improved accuracy in the process meant that components were capable of being built to the tolerances required for assembly purposes (Fit). Improved material properties meant that parts could be properly handled so that they could be assessed according to how they would eventually work (Function).

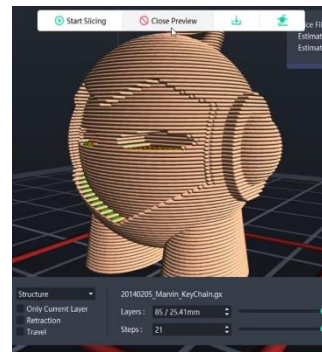
The Generic AM Process

Step 1: CAD

All AM parts must start from a software model that fully describes the external geometry. use of almost any professional CAD solid modeling software, but the output must be a 3D solid or surface representation

Step 2: Conversion to STL

STL file describes the external closed surfaces of the original CAD model and forms the basis for calculation of the slices.



STL (Stereolithography) is a file format native to the stereolithography CAD software created by 3D Systems. STL has several after-the-fact backronyms such as "Standard Triangle Language" and "Standard Tessellation Language".

Step 3: Transfer to AM Machine and STL File Manipulation

The STL file describing the part must be transferred to the AM machine. Here, there may be some general manipulation of the file so that it is the correct size, position, and orientation for building.



Step 4: Machine Setup

The AM machine must be properly set up prior to the build process. Such settings build parameters like the material constraints, energy source, layer thickness, timings, etc.

Step 5: Build

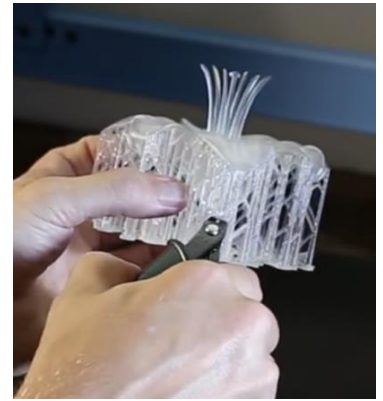
Building the part is mainly an automated process. Only superficial monitoring to ensure no errors have taken place like running out of material, power or software glitches, etc.

Step 6: Removal

Once the build has completed, the parts must be removed. This may require interaction with the machine

Step 7: Post-processing

Parts may require an amount of additional cleaning up. Parts may be weak at this stage or they may have supporting features that must be removed. This therefore often requires time and careful, experienced manual manipulation.



Step 8: Application

Parts may require priming and painting to give an acceptable surface texture and finish. They may also be required to be assembled together with other mechanical or electronic components to form a final model or product.

Distinction between AM and CNC Machining

Material

AM technology was originally developed around polymeric materials, waxes, and paper laminates. When using CNC machining to make final products, it works particularly well for hard, relatively brittle materials like steels and other metal alloys to produce high accuracy parts with well-defined properties.

Some AM parts, in contrast, may have voids or anisotropy that are a function of part orientation, process parameters or how the design was input to the machine, whereas CNC parts will normally be more homogeneous and predictable in quality.

Speed

High-speed CNC machining can generally remove material much faster than AM machines can add a similar volume of material. However, this is only part of the picture, as AM technology can be used to produce a part in a single stage. CNC machines require considerable setup and process planning, particularly as parts become more complex in their geometry.

Complexity

If CNC is being used to create a part directly in a single piece, then there may be some geometric features that cannot be fabricated. Since a machining tool must be carried in a spindle, there may be certain accessibility constraints or clashes preventing the tool from being located on the machining surface of a part. AM processes are not constrained in the same way and undercuts and internal features can be easily built without specific process planning. Certain parts cannot be fabricated by CNC unless they are broken up into components and reassembled at a later stage.

Geometry

AM machines essentially break up a complex, 3D problem into a series of simple 2D cross-sections with a nominal thickness. Since this cannot be easily done in CNC, machining of

surfaces must normally be generated in 3D space.

Programming

Determining the program sequence for a CNC machine can be very involved, including tool selection, machine speed settings, approach position and angle, etc. Many AM machines also have options that must be selected, but the range, complexity, and implications surrounding their choice are minimal in comparison. Incorrect programming of a CNC machine could result in severe damage to the machine and may even be a human safety risk.

Advantages of additive manufacturing processes

Design

Other manufacturing processes require a careful and detailed analysis of the part geometry to determine things like the order in which different features can be fabricated, what tools and processes must be used, and what additional fixtures may be required to complete the part. In contrast,

AM needs only some basic dimensional details and a small amount of understanding as to how the AM machine works and the materials that are used to build the part.

Planning

1. There is no need to define a blank geometry as no blanks are required. 2. Minimum operation or process planning. GMP's are based on one operation only. 3. No complicated scheduling and routing problem. The part is made in one set up.

Tooling/Jig- Fixtures

GMP's are tool-less processes and the whole complex tasks of tool selection and tool management are not needed. No clamping required. Complex jigs and fixtures are also eliminated. Mould dies and tool design, die design are eliminated.

Automation

Introduction of flexible automation is very convenient as the whole shape generation process is generalized in character. The shape generation is done using a computer and interfacing the manufacturing unit with the whole system is very simple.

Classification of the additive manufacturing processes

State of material	Type	Mechanism	Energy type	Energy source	Process
	Photo-polymers	Liquid photo-polymerization	Monochromatic light	Lamp	Solid ground curing (SGC)
				Laser beam	Stereolithography (STL)
			Light(two frequencies)	Two laser beams	Beam interference solidification (BIS)
	Thermo	Liquid thermal	Heat	Laser beam	Thermal stereolithography (TSTL)
	Non-Metals	Melting and solidification	Heat	Heated nozzle	Fused deposition modelling (FDM)
					Ballistic particle manufacturing (BPM)
	Metals	Melting and solidification	Heat	Electric arc	Shape melting
				Laser beam	Fused deposition modelling (FDM)
				Electrochemical discharge	Fused deposition modelling (FDM)
Solid	Thin sheets and foils	Selective gluing and cutting	Adhesive bonding	Glue and laser beam	Laminated Object manufacturing (LOM)
	Semi-polymerized plastic foils	Foil polymerization	Light	Lamp	Solid foil polymerization (SFP)
Powder	Single component	Selective sintering	Heat	laser beam	Selective laser sintering (SLS)
	Coated powder	Selective sintering	Heat	laser beam	Selective laser sintering (SLS)
	One component and one binder	Selective powder binding	Chemical bond	Fine droplet beam of binder liquid	3D-printing or, MIT process, or, selective powder binding (SPB)